2019 Bored of Studies Trial Examinations

Mathematics Extension 2

Solutions

Section I

Answers

1	(C)	**6**	(B)
2	(D)	**7**	(C)
3	(B)	**8**	(B)
4	(D)	**9**	(C)
5	(A)	**10**	(A)

Brief explanations

1 Using the product rule $\dfrac{d^2x}{dy^2} = y + x\dfrac{dy}{dx}$. Substituting $\dfrac{dy}{dx} = xy$ gives the answer (C).

2 Since gravitational force and resistance force are in opposing directions, the net force must be $mg - kv$ or $-mg + kv$ depending on where the positive direction is defined. The equivalent acceleration equation which satisfies any of these is (D).

3 By De-Moivre's theorem the argument of z^k is $\frac{(2m-1)k\pi}{n}$. When $k = 1$, the argument is $\frac{(2m-1)\pi}{n}$. Therefore, the argument between adjacent points must be $\frac{(2m-1)\pi}{n}$ as k is increases. In order to have a full revolution in rotation (before crossing over to the same arguments) k must increase up to $2n$ (i.e. the argument passes an even multiple of $/pi$). Hence, the answer is (B).

4 Options (A) and (C) are not correct because they result in odd functions and the graph has no symmetry. Choosing (B) results in $y = 1 - \frac{1}{x}$, which is a shifted hyperbola and does not match the shape of the graph. Note that when $x \to -\infty$ then $y \to \infty$, which matches the answer (D).

5 Since $\ddot{\theta} = 1$ then $\dot{\theta} = t$ after applying the initial conditions. Integrating again with respect to t gives $\theta = \frac{1}{2}t^2$ after applying initial conditions. Since $r = 1$, the equations in y are given by

$$\begin{aligned} y &= \sin\theta \\ &= \sin\left(\frac{t^2}{2}\right) \\ \dot{y} &= t\cos\left(\frac{t^2}{2}\right) \\ \ddot{y} &= -t^2\sin\left(\frac{t^2}{2}\right) + \cos\left(\frac{t^2}{2}\right) \\ &= \cos\theta - 2\theta\sin\theta \end{aligned}$$

Hence, the answer is (A).

6 Since $0 < \arg z_1 < \frac{\pi}{2}$ and $-\pi < \arg z_2 < -\frac{\pi}{2}$ then $\frac{\pi}{2} < \arg z_1 - \arg z_2 < \frac{3\pi}{2}$. Hence, $\arg\left(\frac{z_1}{z_2}\right)$ must lie in the second and third quadrants, so the answer is (B).

7 This can be visualised with a sketch of the ellipse and shifted rectangular hyperbola. The right branch of the hyperbola intersects the ellipse twice whereas the left branch touches the ellipse at $(-3, 0)$. Hence, there are three points of intersection, so the answer is (C).

8 Creating a common denominator and equating the numerators for each of the options gives

For option (A) $P(x) = a(x+m)(x^2-n^2) + b(x^2-n^2) + c(x+m)^2$
For option (B) $P(x) = a(x^2-n^2) + b(x+m)^2(x-n) + c(x+m)^2(x+n)$
For option (C) $P(x) = a(x^2-n^2) + b(x+m)^2(x-n) + c(x+m)^2$
For option (D) $P(x) = a(x^2-n^2) + b(x+m)^2(x+n) + c(x+m)^2$

Since $P(x)$ is of degree 2 or less, the coefficient of x^3 on the right hand side must be zero. However, for all options except (B) this leads to $a = 0$ or $b = 0$, which contradicts the requirement that they need to be non-zero. For option (B) the required condition is $b + c = 0$ which is possible for non-zero values of b and c. Hence, the answer is (B).

9 The substitution $x = \frac{\pi}{2} - u$ converts the integral to

$$\int_0^{\frac{\pi}{2}} \frac{1}{1+\cos x}\,dx$$

But since $\dfrac{1}{1+\cos x} < \dfrac{1+\sin x}{1+\cos x}$ for $0 < x < \frac{\pi}{2}$ then option (A) cannot be correct.

Expanding the integrand in option (B) gives

$$\begin{aligned} \frac{1}{1+\left(\sin\frac{x}{2}+\cos\frac{x}{2}\right)^2} &= \frac{1}{2+2\sin\frac{x}{2}\cos\frac{x}{2}} \\ &= \frac{1}{2+\sin x} \end{aligned}$$

Since $\dfrac{1}{2+\sin x} < \dfrac{1}{1+\sin x}$ then option (B) cannot be correct.

Note that the integral is positive in the domain $0 < x < \frac{\pi}{2}$ so option (D) cannot be correct because that integrand is actually negative in its domain $0 < x < 1$.

For option (C), note that the integral can be obtained via t-formula substitution $t = \tan\frac{x}{2}$.

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \frac{1}{1+\sin x}\,dx &= \int_0^1 \frac{1}{1+\frac{2t}{1+t^2}} \times \frac{2}{1+t^2}\,dt \\ &= \int_0^1 \frac{2}{(1+t)^2}\,dt \\ &= \int_0^1 \frac{2}{(1+x)^2}\,dx \end{aligned}$$

Hence, the answer is (C).

10 By summing all n equations the result is

$$\begin{aligned} (n+1)x_1 + (n+1)x_2 + (n+1)x_3 + \ldots + (n+1)x_n &= 1+2+3+\ldots+n \\ (n+1)(x_1+x_2+x_3+\ldots+x_n) &= \frac{n(n+1)}{2} \\ x_1+x_2+x_3+\ldots+x_n &= \frac{n}{2} \end{aligned}$$

But

$$\begin{aligned} x_1+x_2+x_3+\ldots+x_{k-1}+2x_k+x_{k+1}+\ldots+x_n &= k \\ x_1+x_2+x_3+\ldots+x_n &= k - x_k \end{aligned}$$

Hence the general solution is $x_k = k - \dfrac{n}{2}$, which is (A).

Question 11

(a) Let $u = \sqrt{x} \Rightarrow dx = 2u\,du$. When $x = 0$ then $u = 1$ and when $x = 1$ then $u = 1$. Apply this substitution to the first fraction and use integration by parts.

$$\begin{aligned}\int_0^1 \frac{\tan^{-1}(\sqrt{x})}{1+x^2}\,dx &= \int_0^1 \tan^{-1} u \times \frac{2u}{1+u^4}\,du \\ &= \int_0^1 \tan^{-1} u \times \frac{d}{du}(\tan^{-1}(u^2))\,du \\ &= [\tan^{-1} u \tan^{-1}(u^2)]_0^1 - \int_0^1 \frac{\tan^{-1}(u^2)}{1+u^2}\,du\end{aligned}$$

$$\int_0^1 \frac{\tan^{-1}(\sqrt{x})}{1+x^2}\,dx + \int_0^1 \frac{\tan^{-1}(u^2)}{1+u^2}\,du = \frac{\pi^2}{16}$$

$$\therefore \int_0^1 \frac{\tan^{-1}(\sqrt{x}) + \tan^{-1}(x^2)}{1+x^2}\,dx = \frac{\pi^2}{16}$$

Alternative solution:

Let $x = \tan\theta \Rightarrow dx = \sec^2\theta\,d\theta$. When $x = 0$ then $\theta = 0$ and when $x = 1$ then $\theta = \frac{\pi}{4}$. Applying this substitution to the whole integral.

$$\begin{aligned}\int_0^1 \frac{\tan^{-1}(\sqrt{x}) + \tan^{-1}(x^2)}{1+x^2}\,dx &= \int_0^{\frac{\pi}{4}} \frac{\tan^{-1}(\sqrt{\tan\theta}) + \tan^{-1}(\tan^2\theta)}{1+\tan^2\theta} \times \sec^2\theta\,d\theta \\ &= \int_0^{\frac{\pi}{4}} \left(\tan^{-1}(\sqrt{\tan\theta}) + \tan^{-1}(\tan^2\theta)\right)\,d\theta\end{aligned}$$

Let $y = \tan^{-1}(\sqrt{\tan x})$ then $x = \tan^{-1}(\tan^2 x)$. The area bounded by the curve $y = \tan^{-1}(\sqrt{\tan x})$, the x-axis and the line $x = \frac{\pi}{4}$ is represented by

$$\int_0^{\frac{\pi}{4}} \tan^{-1}(\sqrt{\tan x})\,dx$$

The area bounded by the curve $y = \tan^{-1}(\sqrt{\tan x})$, the y-axis and the line $y = \frac{\pi}{4}$ (which occurs at $x = \frac{\pi}{4}$) is represented by

$$\int_0^{\frac{\pi}{4}} \tan^{-1}(\tan^2 x)\,dx$$

The sum of the two integrals is the area of the square covering the domain $0 \le x \le \frac{\pi}{4}$ and range $0 \le y \le \frac{\pi}{4}$. Hence

$$\int_0^1 \frac{\tan^{-1}(\sqrt{x}) + \tan^{-1}(x^2)}{1+x^2}\,dx = \frac{\pi^2}{16}$$

(b) (i) Since $P(x)$ has real coefficients then non-real roots occur in conjugate pairs, so $\bar{\alpha}$ is also a root of multiplicity r. Hence, $P(x)$ can be expressed as

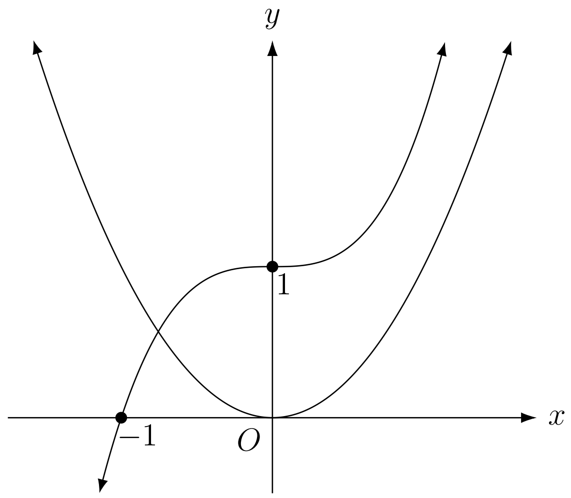
$$\begin{aligned} P(x) &= (x-\alpha)^r(x-\bar{\alpha})^r Q(x) \\ &= (x^2-(\alpha+\bar{\alpha})x+\alpha\bar{\alpha})^r Q(x) \end{aligned}$$

Noting that $(\alpha+\bar{\alpha})$ and $\alpha\bar{\alpha}$ are real.

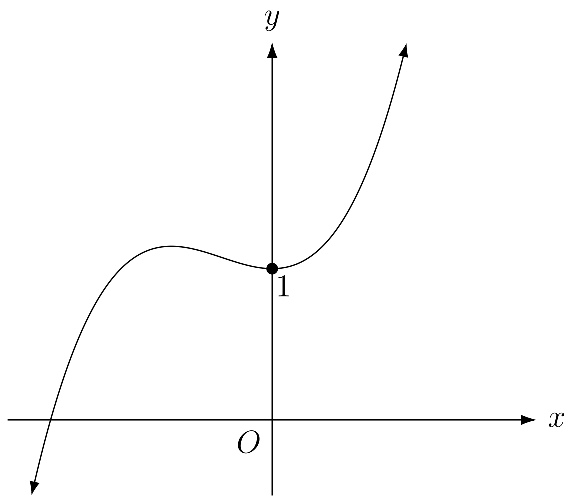
(ii) Using the product rule

$$\begin{aligned} P'(x) &= r(x^2-(\alpha+\bar{\alpha})x+\alpha\bar{\alpha})^{r-1}(2x-(\alpha+\bar{\alpha}))Q(x)+(x^2-(\alpha+\bar{\alpha})x+\alpha\bar{\alpha})^r Q'(x) \\ P'(\alpha) &= r(\alpha^2-(\alpha+\bar{\alpha})\alpha+\alpha\bar{\alpha})^{r-1}(2\alpha-(\alpha+\bar{\alpha}))Q(\alpha)+(\alpha^2-(\alpha+\bar{\alpha})\alpha+\alpha\bar{\alpha})^r Q'(\alpha) \\ &= r(\alpha^2-\alpha^2-\bar{\alpha}\alpha+\alpha\bar{\alpha})^{r-1}(\alpha-\bar{\alpha})Q(\alpha)+(\alpha^2-\alpha^2-\bar{\alpha}\alpha+\alpha\bar{\alpha})^r Q'(\alpha) \\ &= 0 \end{aligned}$$

(c) (i) Sketching the graphs $y=x^3+1$ and $y=x^2$



(ii) Using addition of ordinates to obtain the graph



(iii) From the sketch, the radius of the cylindrical shell is $-x$ because $x < 0$ and the height is y. Hence the volume is

$$\begin{aligned} V &= 2\pi \int_{-1}^{0} (-x)y\,dx \\ &= 2\pi \int_{-1}^{0} (-x^4 - x^3 - x)\,dx \\ &= -2\pi \left[\frac{x^5}{5} + \frac{x^4}{4} + \frac{x^2}{2}\right]_{-1}^{0} \\ &= -2\pi \left(0 - \left(-\frac{1}{5} + \frac{1}{4} + \frac{1}{2}\right)\right) \\ &= \frac{11\pi}{10} \quad \text{cubic units} \end{aligned}$$

(d) Since α, β, γ and δ are the roots of $P(x)$ then by the factor theorem

$$\begin{aligned} P(x) &= (x-\alpha)(x-\beta)(x-\gamma)(x-\delta) \\ P(i) &= (i-\alpha)(i-\beta)(i-\gamma)(i-\delta) \\ &= (\alpha-i)(\beta-i)(\gamma-i)(\delta-i) \\ \text{similarly} \quad P(-i) &= (-i-\alpha)(-i-\beta)(-i-\gamma)(-i-\delta) \\ &= (\alpha+i)(\beta+i)(\gamma+i)(\delta+i) \\ P(i)P(-i) &= (\alpha+i)(\beta+i)(\gamma+i)(\delta+i)(\alpha-i)(\beta-i)(\gamma-i)(\delta-i) \\ &= (\alpha^2+1)(\beta^2+1)(\gamma^2+1)(\delta^2+1) \\ \text{but} \quad P(i) &= i^4 - 3i^3 - 4i^2 + 9i + 2 \\ &= 7 + 12i \\ \text{and} \quad P(-i) &= 7 - 12i \quad \text{as } P \text{ has real coefficients.} \\ \therefore (\alpha^2+1)(\beta^2+1)(\gamma^2+1)(\delta^2+1) &= (7+12i)(7-12i) \\ &= 193 \end{aligned}$$

(e) When circles touch, the line of centres passes through the point of contact, hence

$$\begin{aligned} AR = a + r \quad &\text{and} \quad BR = b + r \\ |AR - BR| &= |a - b| \end{aligned}$$

The points A and B are fixed. The radii a and b are also fixed. As r varies and R moves, the difference in distances AR and BR remains constant. This is analagous to the relationship $|PS - PS'|$ being constant for a variable point P and fixed foci S and S'. So the locus has the shape of a hyperbola where A and B are the foci.

Question 12

(a) (i) Resolving the forces horizontally and vertically

$$N\sin\theta + F\cos\theta = \frac{mv^2}{r}$$
$$N\cos\theta - F\sin\theta = mg$$
$$N(\sin^2\theta + \cos^2\theta) + F\sin\theta\cos\theta - F\sin\theta\cos\theta = \frac{mv^2\sin\theta}{r} + mg\cos\theta$$
$$\Rightarrow N = \frac{mv^2\sin\theta}{r} + mg\cos\theta$$
$$N\sin\theta\cos\theta - N\cos\theta\sin\theta + F(\cos^2\theta + \sin^2\theta) = \frac{mv^2\cos\theta}{r} - mg\sin\theta$$
$$\Rightarrow F = \frac{mv^2\cos\theta}{r} - mg\sin\theta$$

Since $F \geq 0$, $m > 0$ and $\cos\theta > 0$ then

$$\frac{mv^2\cos\theta}{r} - mg\sin\theta \geq 0$$
$$v^2 \geq rg\tan\theta$$

(ii) Since $F \leq \mu N$, $m > 0$ and $\cos\theta > 0$ then

$$\frac{mv^2\cos\theta}{r} - mg\sin\theta \leq \mu\left(\frac{mv^2\sin\theta}{r} + mg\cos\theta\right)$$
$$\frac{v^2}{r}(\cos\theta - \mu\sin\theta) \leq g(\sin\theta + \mu\cos\theta)$$
$$v^2(1 - \mu\tan\theta) \leq rg(\tan\theta + \mu)$$

For $v^2 \leq \dfrac{rg(\tan\theta + \mu)}{1 - \mu\tan\theta}$ to hold, then it must be that $1 - \mu\tan\theta > 0$, otherwise the inequality sign will be switched. Thus, the range of values allowed for μ are those where $\mu < \cot\theta$.

(b) (i) The equation of a general normal at $\left(ct, \frac{c}{t}\right)$ is

$$\begin{aligned} t^3x - ty - c(t^4-1) &= 0 \\ ct^4 - t^3x + ty - c &= 0 \end{aligned}$$

This is a quartic polynomial in t which has the roots t_1, t_2, t_3 and t_4. Since $R(a,b)$ lies on all four normals then the polynomial equation is

$$ct^4 - t^3a + tb - c = 0$$

By product of the roots

$$\begin{aligned} t_1t_2t_3t_4 &= \frac{-c}{c} \\ &= -1 \end{aligned}$$

(ii) First consider

$$\begin{aligned} \frac{1}{t_1}+\frac{1}{t_2}+\frac{1}{t_3}+\frac{1}{t_4} &= \frac{t_2t_3t_4+t_1t_3t_4+t_1t_2t_4+t_1t_2t_3}{t_1t_2t_3t_4} \\ &= \frac{-\frac{b}{c}}{-1} \\ &= \frac{b}{c} \end{aligned}$$

Also, using the result in (i)

$$\begin{aligned} \frac{1}{t_1t_2}+\frac{1}{t_1t_3}+\frac{1}{t_1t_4}+\frac{1}{t_2t_3}+\frac{1}{t_2t_4}+\frac{1}{t_3t_4} &= -t_3t_4-t_2t_4-t_2t_3-t_1t_4-t_1t_3-t_1t_2 \\ &= 0 \end{aligned}$$

But

$$\left(\frac{1}{t_1}+\frac{1}{t_2}+\frac{1}{t_3}+\frac{1}{t_4}\right)^2 = \frac{1}{t_1^2}+\frac{1}{t_2^2}+\frac{1}{t_3^2}+\frac{1}{t_4^2}+2\left(\frac{1}{t_1t_2}+\frac{1}{t_1t_3}+\frac{1}{t_1t_4}+\frac{1}{t_2t_3}+\frac{1}{t_2t_4}+\frac{1}{t_3t_4}\right)$$

$$\begin{aligned} \frac{1}{t_1^2}+\frac{1}{t_2^2}+\frac{1}{t_3^2}+\frac{1}{t_4^2} &= \left(\frac{b}{c}\right)^2 - 2\times 0 \\ &= \frac{b^2}{c^2} \end{aligned}$$

(iii) Consider

$$
\begin{aligned}
(P_kR)^2 &= (ct_k - a)^2 - \left(\frac{c}{t_k} - b\right)^2 \\
&= c^2t_k^2 - 2act_k + a^2 + \frac{c^2}{t_k^2} - \frac{2bc}{t_k} + b^2
\end{aligned}
$$

Using the result in (ii)

$$
\begin{aligned}
\sum_{k=1}^{4}(P_kR)^2 &= c^2\sum_{k=1}^{4} t_k^2 - 2ac\sum_{k=1}^{4} t_k + 4a^2 + c^2\sum_{k=1}^{4}\frac{1}{t_k^2} - 2bc\sum_{k=1}^{4}\frac{1}{t_k} + 4b^2 \\
&= c^2\sum_{k=1}^{4} t_k^2 - 2ac \times \frac{a}{c} + 4a^2 + c^2 \times \frac{b^2}{c^2} - 2bc \times \frac{b}{c} + 4b^2 \\
&= c^2\sum_{k=1}^{4} t_k^2 + 2a^2 + 3b^2
\end{aligned}
$$

But

$$
\begin{aligned}
\sum_{k=1}^{4} t_k^2 &= \left(\sum_{k=1}^{4} t_k\right)^2 - 2(t_3t_4 + t_2t_4 + t_2t_3 + t_1t_4 + t_1t_3 + t_1t_2) \\
&= \left(\frac{a}{c}\right)^2 - 2 \times 0 \\
&= \frac{a^2}{c^2}
\end{aligned}
$$

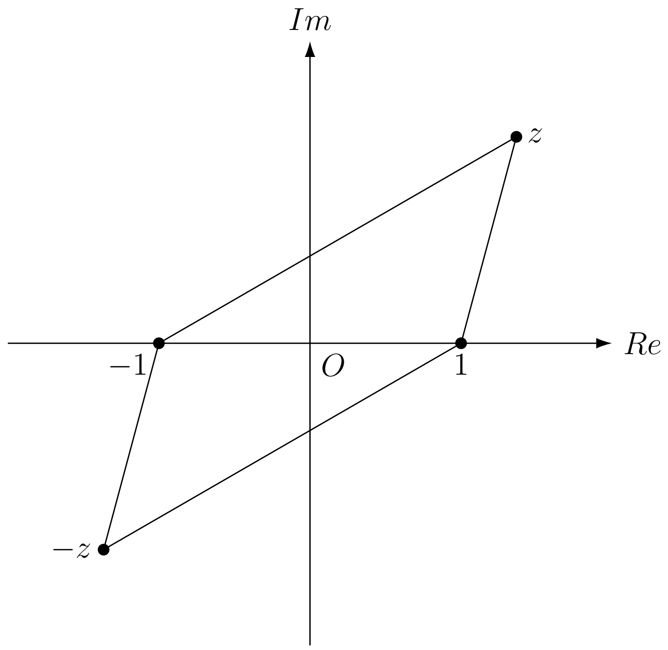
Hence

$$
\begin{aligned}
\sum_{k=1}^{4}(P_kR)^2 &= 3a^2 + 3b^2 \\
\therefore \frac{(P_1R)^2 + (P_2R)^2 + (P_3R)^2 + (P_4R)^2}{3} &= a^2 + b^2
\end{aligned}
$$

(c) (i) Using the cosine rule on $\triangle ABC$ and $\triangle BCD$

$$
\begin{aligned}
AC^2 &= AB^2 + BC^2 - 2(AB)(BC)\cos\angle ABC \\
BD^2 &= CD^2 + BC^2 - 2(CD)(BC)\cos\angle BCD \\
\text{But}\quad AB &= CD \quad \text{(opposite sides of parallelogram)} \\
\angle BCD + \angle ABC &= \pi \quad \text{(adjacent angles of parallelogram)} \\
\Rightarrow BD^2 &= AB^2 + BC^2 - 2(AB)(BC)\cos(\pi - \angle ABC) \\
&= AB^2 + BC^2 + 2(AB)(BC)\cos\angle ABC \\
\therefore AC^2 + BD^2 &= 2(AB^2 + AD^2)
\end{aligned}
$$

(ii) Labelling the points on the Argand diagram



By addition and subtraction of vectors it can be seen that

$$
\begin{aligned}
|\overrightarrow{AB}| &= |z+1| \\
|\overrightarrow{AD}| &= |z-1| \\
|\overrightarrow{AC}| &= 2|z| \\
|\overrightarrow{BD}| &= 2
\end{aligned}
$$

Substitute the lengths into the result in (i)

$$4|z|^2 + 4 = 2(|z+1|^2 + |z-1|^2)$$

But the locus is given by

$$
\begin{aligned}
\left|z - \frac{1}{z}\right| &= 2 \\
\Rightarrow |z^2 - 1| &= 2|z| \quad \text{noting that } z \neq 0
\end{aligned}
$$

Substitute this into the result

$$
\begin{aligned}
|z^2-1|^2 - 2|z+1|^2 - 2|z-1|^2 + 4 &= 0 \\
|z+1|^2(|z-1|^2 - 2) - 2(|z-1|^2 - 2) &= 0 \\
(|z+1)^2 - 2)(|z-1|^2 - 2) &= 0
\end{aligned}
$$

The locus of z is the set of points lying on $|z+1|^2 = 2$ or $|z-1|^2 = 2$.
This represents two circles of equal radii $\sqrt{2}$ with centres $(-1, 0)$ and $(1, 0)$.

Question 13

(a) (i) Since $F_1 = m_1\ddot{x}_1$ and $F_2 = m_2\ddot{x}_2$ then

$$\begin{aligned} m_1\ddot{x}_1 &= \frac{Gm_1m_2}{r^2} \\ \Rightarrow \ddot{x}_1 &= \frac{Gm_2}{r^2} \\ m_2\ddot{x}_2 &= -\frac{Gm_1m_2}{r^2} \\ \Rightarrow \ddot{x}_2 &= \frac{Gm_1}{r^2} \end{aligned}$$

(ii) Since $r = x_2 - x_1$ and $M = m_1 + m_2$ then

$$\begin{aligned} \ddot{r} &= \ddot{x}_2 - \ddot{x}_1 \\ &= -\frac{Gm_2}{r^2} - \frac{Gm_1}{r^2} \\ &= -\frac{G(m_1 + m_2)}{r^2} \\ &= -\frac{GM}{r^2} \end{aligned}$$

(iii) Using $\ddot{r} = \dfrac{d}{dr}\left(\dfrac{\dot{r}^2}{2}\right)$

$$\begin{aligned} \frac{\dot{r}^2}{2} &= -\frac{GM}{r} + c \quad \text{when} \quad r = r_0, \quad \dot{r} = \dot{x}_2 - \dot{x}_1 = 0 \Rightarrow c = -\frac{GM}{r_0} \\ \dot{r}^2 &= 2GM\left(\frac{1}{r} - \frac{1}{r_0}\right) \\ \dot{r} &= -\sqrt{2GM\left(\frac{1}{r} - \frac{1}{r_0}\right)} \quad \text{since } r \text{ is decreasing over time so } \dot{r} < 0 \end{aligned}$$

(iv) Use the substitution $\theta = \sin^{-1}\sqrt{\dfrac{r}{r_0}} \Rightarrow r = r_0 \sin^2\theta \Rightarrow dr = 2r_0 \sin\theta\cos\theta\, d\theta$

$$\begin{aligned}
\frac{dr}{dt} &= -\sqrt{2GM\left(\frac{1}{r} - \frac{1}{r_0}\right)} \\
\frac{dt}{dr} &= -\sqrt{\frac{r_0}{2GM}} \times \sqrt{\frac{r}{r_0 - r}} \\
t &= -\sqrt{\frac{r_0}{2GM}} \int \sqrt{\frac{r}{r_0 - r}}\, dr \\
&= -\sqrt{\frac{r_0}{2GM}} \int \sqrt{\frac{r_0 \sin^2\theta}{r_0(1 - \sin^2\theta)}} \times 2r_0 \sin\theta\cos\theta\, d\theta
\end{aligned}$$

But the range implied by the substitution for $0 < r < r_0$ is $0 < \theta < \frac{\pi}{2}$ so $\sin\theta$ and $\cos\theta$ are all positive. Hence

$$\begin{aligned}
t &= -\sqrt{\frac{r_0}{2GM}} \int \frac{\sin\theta}{\cos\theta} \times 2r_0 \sin\theta\cos\theta\, d\theta \\
&= \frac{r_0\sqrt{r_0}}{\sqrt{2GM}} \int (-2\sin^2\theta)\, d\theta \\
&= \frac{r_0\sqrt{r_0}}{\sqrt{2GM}} \int (\cos 2\theta - 1)\, d\theta \\
&= \frac{r_0\sqrt{r_0}}{\sqrt{2GM}} \left(\frac{1}{2}\sin 2\theta - \theta\right) + C
\end{aligned}$$

When $t \to 0^+$ and $r \to r_0^-$ then $\theta \to \frac{\pi}{2}^-$ so $C = \dfrac{r_0\sqrt{r_0}}{\sqrt{2GM}} \times \dfrac{\pi}{2}$

$$\begin{aligned}
t &= \frac{r_0\sqrt{r_0}}{\sqrt{2GM}} \left(\frac{1}{2}\sin 2\theta - \theta\right) + \frac{r_0\sqrt{r_0}}{\sqrt{2GM}} \times \frac{\pi}{2} \\
&= \frac{r_0\sqrt{r_0}}{\sqrt{2GM}} \left(\frac{\pi}{2} + \frac{1}{2}\sin 2\theta - \theta\right)
\end{aligned}$$

(v) Since $\sin x < x$ for $x > 0$ then $\sin 2\theta < 2\theta$ for $0 < \theta < \frac{\pi}{2}$ as defined in part (iv). This implies that

$$\begin{aligned}
t &< \frac{r_0\sqrt{r_0}}{\sqrt{2GM}} \left(\frac{\pi}{2} + \frac{1}{2} \times 2\theta - \theta\right) \\
&= \frac{\pi r_0\sqrt{r_0}}{2\sqrt{2GM}} \\
&= \frac{\pi}{\sqrt{GM}} \left(\frac{r_0}{2}\right)^{\frac{3}{2}}
\end{aligned}$$

The upper bound of t suggests that this system of P_1 and P_2 moving towards each other cannot be sustained beyond time $t = \dfrac{\pi}{\sqrt{GM}}\left(\dfrac{r_0}{2}\right)^{\frac{3}{2}}$ (in fact, they collide at that time).

(b) From the geometric series

$$\begin{aligned}\frac{(1+x)^n-1}{(1+x)-1} &= 1+(1+x)+(1+x)^2+...+(1+x)^{n-1}\\ \Rightarrow (1+x)^n-1 &= x\left[1+(1+x)+(1+x)^2+...+(1+x)^{n-1}\right]\\ &= \sum_{k=1}^{n} x(1+x)^{k-1}\end{aligned}$$

When $x>0$ then

$$\begin{aligned}(1+x)^{k-1} &> 1\\ \Rightarrow x(1+x)^{k-1} &> x\end{aligned}$$

When $-1 \leq x \leq 0$ then

$$\begin{aligned}(1+x)^{k-1} &\leq 1\\ \Rightarrow x(1+x)^{k-1} &\geq x\end{aligned}$$

Hence

$$\begin{aligned}(1+x)^n-1 &\geq \sum_{k=1}^{n} x\\ \therefore (1+x)^n &\geq 1+nx\end{aligned}$$

(c) (i) Let $x=\dfrac{a_{n+1}}{a_n}-1$ and replace n with $n+1$ on the result in part (b)

$$\begin{aligned}\left(\frac{a_{n+1}}{a_n}\right)^{n+1} &\geq 1+(n+1)\left(\frac{a_{n+1}}{a_n}-1\right)\\ &= 1+(n+1)\times\frac{x_1+x_2+...+x_{n+1}}{n+1}\times\frac{n}{x_1+x_2+...+x_n}-(n+1)\\ &= \frac{n(x_1+x_2+...+x_n)+nx_{n+1}}{x_1+x_2+...+x_n}-n\\ &= \frac{nx_{n+1}}{x_1+x_2+...+x_n}\\ &= \frac{x_{n+1}}{a_n}\end{aligned}$$

(ii) For $n = 1$, LHS $= x_1$ and RHS $= \left(\frac{x_1}{1}\right)^1 = x_1$ $\quad\therefore$ Statement true for $n = 1$
Assume the statement is true for $n = k$

$$x_1 x_2 ... x_k \le \left(\frac{x_1 + x_2 + ... + x_k}{k}\right)^k$$

Required to prove the statement is true for $n = k + 1$

$$x_1 x_2 ... x_{k+1} \le \left(\frac{x_1 + x_2 + ... + x_{k+1}}{k+1}\right)^{k+1}$$

$$\begin{aligned}
\text{LHS} &= x_1 x_2 ... x_k x_{k+1} \\
&\le \left(\frac{x_1 + x_2 + ... + x_k}{k}\right)^k x_{k+1} \quad \text{by assumption} \\
&= \left(\frac{x_1 + x_2 + ... + x_k}{k}\right)^k x_{k+1} \times \frac{a_k}{a_k} \\
&= \frac{x_{k+1}}{a_k} \times a_k^{k+1} \\
&\le \left(\frac{a_{k+1}}{a_k}\right)^{k+1} \times a_k^{k+1} \quad \text{from part (i)} \\
&= a_{k+1}^{k+1} \\
&= \left(\frac{x_1 + x_2 + ... + x_{k+1}}{k+1}\right)^{k+1} \\
&= \text{RHS}
\end{aligned}$$

Hence the statement is true by induction for positive integers n.

Question 14

(a) (i) From the relationship between the axes and the eccentricity, noting that $e > 1$ for the hyperbola

$$\begin{aligned} 3a^2 &= a^2(e^2 - 1) \\ e^2 &= 4 \\ \therefore e &= 2 \end{aligned}$$

(ii) Since $e = 2$ then the coordinates of S are $(2a, 0)$ and the equation of the directrix is $x = \frac{a}{2}$.

The coordinates of X are $(-a, 0)$, this means that the midpoint of XS lies on the directrix. Hence, X and S are in fact reflections of each other about the positive directrix.

Since P' and P are also reflections of each other about the positive directrix then $XP' = SP$.

Also, $CP = CP'$ and $XC = SC$ due to equal radii.

$$\therefore \triangle SCP \equiv \triangle XCP' \quad (SSS)$$

(iii) Let M be a point on the directrix where it is the midpoint of PP'. Since P' is a reflection of P about the positive directrix then PM is perpendicular to the directrix. Using the locus definition of the hyperbola and the result in part (i)

$$\begin{aligned} \frac{PS}{PM} &= 2 \\ PS &= 2PM \qquad \text{but } PP' = 2PM \\ \Rightarrow PS &= PP' \\ PC & \text{ is common} \\ P'C &= PC \quad \text{(equal radii)} \\ \therefore \triangle PCP' &\equiv \triangle SCP \quad (SSS) \end{aligned}$$

This means that $\triangle XCP'$, $\triangle P'CP$ and $\triangle PCS$ are all congruent, which implies $\angle XCP' = \angle P'CP = \angle PCS$. But

$$\begin{aligned} \angle SCX &= \angle PCX' + \angle PCP' + \angle SCP \\ \therefore \angle SCX &= 3 \times \angle PCP' \end{aligned}$$

(b) (i) Let $y = \dfrac{1}{x} \Rightarrow x = \dfrac{1}{y}$ and substitute into the polynomial equation

$$\left(\frac{1}{y}\right)^4 - a\left(\frac{1}{y}\right)^3 - b\left(\frac{1}{y}\right)^2 - c\left(\frac{1}{y}\right) - d = 0$$
$$dy^4 + cy^3 + by^2 + ay - 1 = 0$$

Hence equation with the roots $\dfrac{1}{\alpha}, \dfrac{1}{\beta}, \dfrac{1}{\gamma}$ and $\dfrac{1}{\delta}$ is

$$dx^4 + cx^3 + bx^2 + ax - 1 = 0$$

(ii) If n is a root of $P(x)$ then $\dfrac{1}{n}$ is a root of the equation in (i) so

$$d\left(\frac{1}{n}\right)^4 + c\left(\frac{1}{n}\right)^3 + b\left(\frac{1}{n}\right)^2 + a\left(\frac{1}{n}\right) - 1 = 0$$

But since the coefficients are all positive integers and n is also a positive integer then

$$d\left(\frac{1}{n}\right)^4 + c\left(\frac{1}{n}\right)^3 + b\left(\frac{1}{n}\right)^2 > 0$$
$$d\left(\frac{1}{n}\right)^4 + c\left(\frac{1}{n}\right)^3 + b\left(\frac{1}{n}\right)^2 + a\left(\frac{1}{n}\right) - 1 > a\left(\frac{1}{n}\right) - 1$$
$$\frac{a}{n} - 1 < 0$$
$$n > a$$

(iii) Since n is a positive integer root and $P(x)$ has integer coefficients then n must be a divisor of the constant term $-d$. Since d is a positive integer then

$$\frac{d}{n} \geq 1$$
$$n \leq d \quad \text{but } a \geq d$$
$$\Rightarrow n \leq a$$

This contradicts the inequality in part (ii) and n cannot satisfy both inequalities at the same time. Hence, the initial assumption that $P(x)$ has a positive integer root cannot be be true.

Alternative solution:

Since $a \geq b \geq c \geq d$ then

$$
\begin{aligned}
d\left(\frac{1}{n}\right)^4 + c\left(\frac{1}{n}\right)^3 + b\left(\frac{1}{n}\right)^2 + a\left(\frac{1}{n}\right) &\leq a\left(\frac{1}{n}\right)^4 + a\left(\frac{1}{n}\right)^3 + a\left(\frac{1}{n}\right)^2 + a\left(\frac{1}{n}\right) \\
1 &\leq \frac{a}{n}\left(1 + \frac{1}{n} + \frac{1}{n^2} + \frac{1}{n^3}\right) \\
&= \frac{a}{n}\left(\frac{1 - \frac{1}{n^4}}{1 - \frac{1}{n}}\right) \\
&< \frac{a}{n}\left(\frac{1}{1 - \frac{1}{n}}\right) \\
\Rightarrow n - 1 &< a \\
\Rightarrow n &< a + 1
\end{aligned}
$$

This inequality and the previous inequality imply that n is an integer lying between two consecutive integers. This is a contradiction, so this means the initial assumption that $P(x)$ has a positive integer root cannot be be true.

(c) (i) Writing the sums in full

$$\sum_{k=1}^{2n-1} \frac{k}{1-\omega^k} = \frac{1}{1-\omega} + \frac{2}{2-\omega^2} + \frac{3}{1-\omega^3} + \frac{4}{1-\omega^4} + \dots \frac{2n-2}{1-\omega^{2n-2}} + \frac{2n-1}{1-\omega^{2n-1}}$$

$$\sum_{k=1}^{n-1} \frac{2k}{1-\omega^{2k}} = \frac{2}{1-\omega^2} + \frac{4}{1-\omega^4} + \frac{6}{1-\omega^6} + \frac{8}{1-\omega^8} + \dots + \frac{2n-4}{1-\omega^{2n-4}} + \frac{2n-2}{1-\omega^{2n-2}}$$

$$
\begin{aligned}
\text{Hence RHS} &= \sum_{k=1}^{2n-1} \frac{k}{1-\omega^k} - \sum_{k=1}^{n-1} \frac{2k}{1-\omega^{2k}} \\
&= \frac{1}{1-\omega} + \frac{3}{1-\omega^3} + \frac{5}{1-\omega^5} + \dots + \frac{2n-1}{1-\omega^{2n-1}} \\
&= \text{LHS}
\end{aligned}
$$

(ii) Consider the general term of the second sum, which can be written as

$$\begin{aligned}\frac{2k}{1-\omega^{2k}} &= \frac{2k}{(1-\omega^k)(1+\omega^k)}\\ &= \frac{k(1-\omega^k+1+\omega^k)}{(1-\omega^k)(1+\omega^k)}\\ &= \frac{k}{1+\omega^k}+\frac{k}{1-\omega^k}\end{aligned}$$

Hence

$$\begin{aligned}\sum_{k=1}^{2n-1}\frac{k}{1-\omega^k}-\sum_{k=1}^{n-1}\frac{2k}{1-\omega^{2k}} &= \sum_{k=1}^{2n-1}\frac{k}{1-\omega^k}-\sum_{k=1}^{n-1}\frac{k}{1+\omega^k}-\sum_{k=1}^{n-1}\frac{k}{1-\omega^k}\\ &= \sum_{k=n}^{2n-1}\frac{k}{1-\omega^k}-\sum_{k=1}^{n-1}\frac{k}{1+\omega^k}\end{aligned}$$

(iii) Writing the first sum out in full and noting that $\omega^n=-1$

$$\begin{aligned}\sum_{k=n}^{2n-1}\frac{k}{1-\omega^k} &= \frac{n}{1-\omega^n}+\frac{n+1}{1-\omega^{n+1}}+\frac{n+2}{1-\omega^{n+2}}+\ldots+\frac{2n-1}{1-\omega^{2n-1}}\\ &= \frac{n}{1+\omega^0}+\frac{n+1}{1+\omega^1}+\frac{n+2}{1+\omega^2}+\ldots+\frac{2n-1}{1+\omega^{n-1}}\\ &= \sum_{k=0}^{n-1}\frac{n+k}{1+\omega^k}\end{aligned}$$

$$\begin{aligned}\sum_{k=n}^{2n-1}\frac{k}{1-\omega^k}-\sum_{k=1}^{n-1}\frac{k}{1+\omega^k} &= \sum_{k=0}^{n-1}\frac{n+k}{1+\omega^k}-\sum_{k=1}^{n-1}\frac{k}{1+\omega^k}+\frac{0}{1+\omega^0}\\ &= \sum_{k=0}^{n-1}\frac{n+k}{1+\omega^k}-\sum_{k=0}^{n-1}\frac{k}{1+\omega^k}\\ &= \sum_{k=0}^{n-1}\frac{n+k-k}{1+\omega^k}\\ &= \sum_{k=0}^{n-1}\frac{n}{1+\omega^k}\end{aligned}$$

Hence, using the result from (i)

$$\frac{1}{n}\left[\frac{1}{1-\omega}+\frac{3}{1-\omega^3}+\ldots+\frac{2n-1}{1-\omega^{2n-1}}\right]=1+\frac{1}{1+\omega}+\frac{1}{1+\omega^2}+\ldots+\frac{1}{1+\omega^{n-1}}$$

Question 15

(a) Consider the fact that

$$\begin{aligned} b^3+c^3 &= (b+c)(b^2-bc+c^2) \\ &= (b+c)((b+c)^2-3bc) \end{aligned}$$

But since a, b and c are the sides of a triangle then $b+c>a$ so

$$\begin{aligned} b^3+c^3 &> a(a^2-3bc) \\ \Rightarrow a^3+b^3+c^3+3abc &> 2a^3 \\ \therefore \sqrt[3]{\frac{a^3+b^3+c^3+3abc}{2}} &> a \end{aligned}$$

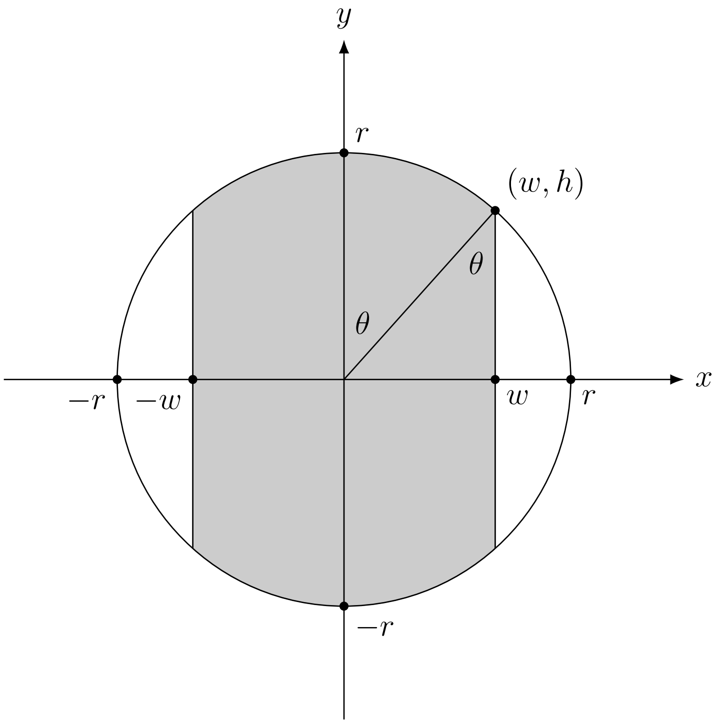
(b) (i) For $n \geq 2$

$$\begin{aligned} I_n + I_{n-2} &= \int_0^{\frac{\pi}{4}} \tan^n x\,dx + \int_0^{\frac{\pi}{4}} \tan^{n-2} x\,dx \\ &= \int_0^{\frac{\pi}{4}} \tan^{n-2} x(\tan^2 x+1)\,dx \\ &= \int_0^{\frac{\pi}{4}} \tan^{n-2} x \sec^2 x\,dx \\ &= \left[\frac{\tan^{n-1} x}{n-1}\right]_0^{\frac{\pi}{4}} \\ &= \frac{1}{n-1} \end{aligned}$$

(ii) Using the result in (i), note that

$$\begin{aligned} I_6+I_4-(I_4+I_2)+(I_2+I_0) &= \frac{1}{5}-\frac{1}{3}+1 \\ I_6+I_0 &= \frac{13}{15} \\ I_6 &= \frac{13}{15} - \int_0^{\frac{\pi}{4}} dx \\ &= \frac{13}{15}-\frac{\pi}{4} \end{aligned}$$

(c) Let θ be the acute angle as shown below in the sector



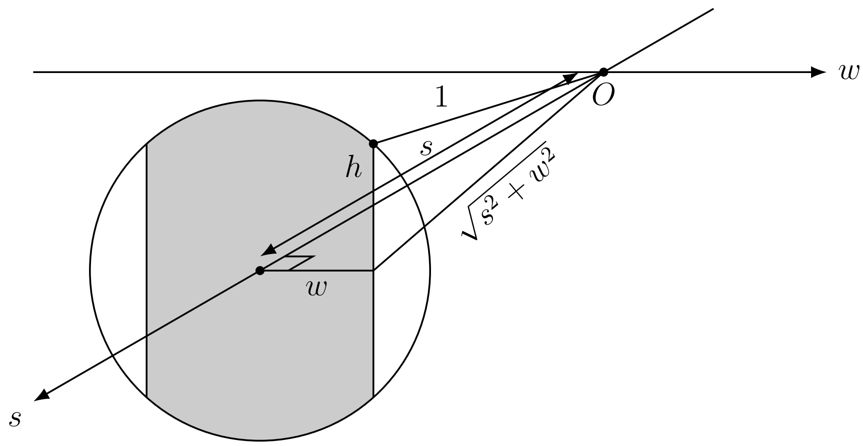
The area of the sector in the first quadrant subtending the angle θ at the centre is $\frac{1}{2}r^2\theta$. The area of the right-angled triangle below it is $\frac{1}{2}wh$. By symmmetry across the four quadrants, the shaded area is given by

$$\begin{aligned} A &= 4 \times \left(\frac{1}{2}r^2\theta + \frac{1}{2}wh\right) \\ &= 2r^2\theta + 2wh \end{aligned}$$

But $\tan\theta = \frac{w}{h}$, or equivalently $\theta = \tan^{-1}\left(\frac{w}{h}\right)$ and $w^2 + h^2 = r^2$ hence

$$A = 2(w^2 + h^2)\tan^{-1}\left(\frac{w}{h}\right) + 2hw$$

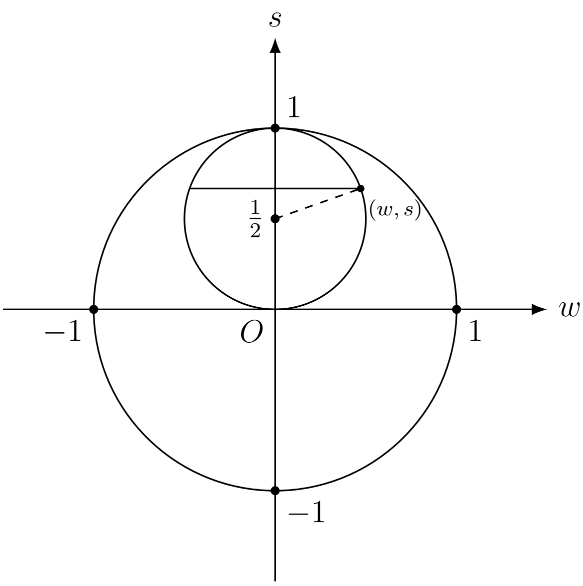
(d) (i) Viewing the slice from the front, it can be seen that s is the distance from the centre of the slice to the centre of the sphere.



The distances h and w are defined similar to part (c). By Pythagoras' theorem, the length of the hypotenuse shown in the diagram to the centre of sphere is $\sqrt{s^2+w^2}$. Since the radius of the sphere is 1 then

$$(\sqrt{s^2+w^2})^2 + h^2 = 1$$
$$\therefore h^2 + s^2 + w^2 = 1$$

(ii) The circular cross-section of the cylinder intersects circular cross-section of the sphere when viewed from above.



Since the radius of the cylinder is $\frac{1}{2}$ and w is half the width of the slice then by Pythagoras' theorem in the cylindrical cross section

$$\left(s - \frac{1}{2}\right)^2 + w^2 = \frac{1}{4}$$

(iii) Making w the subject from part (ii), noting that $w > 0$

$$\begin{aligned} w^2 &= \frac{1}{4} - \left(s - \frac{1}{2}\right)^2 \\ &= \frac{1}{4} - s^2 + s - \frac{1}{4} \\ &= s(1-s) \\ w &= \sqrt{s(1-s)} \end{aligned}$$

Making h the subject from part (i), noting that $h > 0$

$$\begin{aligned} h^2 &= 1 - s^2 - w^2 \\ &= 1 - s^2 - s(1-s) \\ &= 1 - s \\ h &= \sqrt{1-s} \end{aligned}$$

From part (c), the area of the slice is given by

$$\begin{aligned} A &= 2(w^2 + h^2)\tan^{-1}\left(\frac{w}{h}\right) + 2hw \\ &= 2(1-s^2)\tan^{-1}\left(\frac{\sqrt{s(1-s)}}{\sqrt{1-s}}\right) + 2 \times \sqrt{1-s} \times \sqrt{s(1-s)} \\ &= 2(1-s^2)\tan^{-1}\sqrt{s} + 2\sqrt{s}(1-s) \end{aligned}$$

(iv) Let $\theta = \tan^{-1}\sqrt{s} \Rightarrow s = \tan^2\theta \Rightarrow ds = 2\tan\theta\sec^2\theta\, d\theta$.
When $s = 0$ then $\theta = 0$ and when $s = 1$ then $\theta = \frac{\pi}{4}$.

$$\begin{aligned} V &= \int_0^1 A\, ds \\ &= 4\int_0^{\frac{\pi}{4}} \theta(1 - \tan^4\theta)\tan\theta\sec^2\theta\, d\theta + 2\int_0^1 \sqrt{s}(1-s)\, ds \\ &= 4I + 2J \end{aligned}$$

where

$$I = \int_0^{\frac{\pi}{4}} \theta(\tan\theta - \tan^5\theta)\sec^2\theta\, d\theta \quad \text{and} \quad J = \int_0^1 \sqrt{s}(1-s)\, ds$$

Evaluating J first

$$\begin{aligned} J &= \int_0^1 (\sqrt{s} - s\sqrt{s})\, ds \\ &= \left[\frac{2s^{\frac{3}{2}}}{3} - \frac{2s^{\frac{5}{2}}}{5}\right]_0^1 \\ &= \frac{4}{15} \end{aligned}$$

Using integration by parts on I

$$
\begin{aligned}
I &= \left[\theta\left(\frac{\tan^2\theta}{2} - \frac{\tan^6\theta}{6}\right)\right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \left(\frac{\tan^2\theta}{2} - \frac{\tan^6\theta}{6}\right) d\theta \\
&= \frac{\pi}{12} - \frac{1}{2}\int_0^{\frac{\pi}{4}} (\sec^2\theta - 1)\, d\theta + \frac{1}{6}\int_0^{\frac{\pi}{4}} \tan^6\theta\, d\theta \\
&= \frac{\pi}{12} - \frac{1}{2}\left[\tan\theta - \theta\right]_0^{\frac{\pi}{4}} + \frac{1}{6}\left(\frac{13}{15} - \frac{\pi}{4}\right) \quad \text{from (b)(ii)} \\
&= \frac{\pi}{12} - \frac{1}{2} + \frac{\pi}{8} + \frac{13}{90} - \frac{\pi}{24} \\
&= \frac{\pi}{6} - \frac{32}{90}
\end{aligned}
$$

Hence

$$
\begin{aligned}
V &= 4 \times \left(\frac{\pi}{6} - \frac{32}{90}\right) + 2 \times \frac{4}{15} \\
&= \frac{2(3\pi - 4)}{9} \quad \text{cubic units}
\end{aligned}
$$

Question 16

(a) (i) A circular cross section can be drawn along the surface of the smaller sphere such that S_1 and P_1 are tangents to this circle which intersect at P_0. Since tangents to an external point are equal in length then $P_0P_1 = P_0S_1$.

(ii) Using a similar argument to part (i), $P_0P_2 = P_0S_2$. This means that

$$P_0P_1 + P_0P_2 = P_0S_1 + P_0S_2$$

However, $P_0P_1 + P_0P_2 = P_1P_2$ which is a fixed length independent of the position of P_0 because it is the distance between the fixed circles $\mathscr{C}_1$ and $\mathscr{C}_2$ along the cone.

This means that $P_0S_1 + P_0S_2$ is a constant. Since S_1 and S_2 are fixed points, then the locus of P_0 is precisely the flat ellipse with foci S_1 and S_2.

(b) (i) The angles of $\triangle ABC$ can be found as follows.

$$\begin{aligned}
&\angle ACB = \alpha + \beta \quad \text{(exterior angle of triangle)}\\
&\angle ABC = \alpha - \beta \quad \text{(cone is right so angle to the horizontal is } \alpha\text{)}\\
&\angle BAC = \pi - 2\alpha \quad \text{(angle sum of triangle)}
\end{aligned}$$

Using the sine rule and noting that BC is the major axis of the ellipse so it has length $2a$

$$\begin{aligned}
\frac{AB}{\sin\angle ACB} &= \frac{BC}{\sin\angle BAC}\\
AB &= \frac{2a\sin(\alpha+\beta)}{\sin(\pi-2\alpha)}\\
&= \frac{2a\sin(\alpha+\beta)}{\sin 2\alpha}
\end{aligned}$$

(ii) Using the sine rule again

$$\begin{aligned}
\frac{AC}{\sin\angle ABC} &= \frac{BC}{\sin\angle BAC}\\
AC &= \frac{2a\sin(\alpha-\beta)}{\sin(\pi-2\alpha)}\\
&= \frac{2a\sin(\alpha-\beta)}{\sin 2\alpha}
\end{aligned}$$

(iii) Note that the radii are perpendicular to the tangent at the points of contact. The area of $\triangle ABC$ can be found using the radii so

$$\text{Area} = \frac{1}{2} \times r \times AB + \frac{1}{2} \times r \times BC + \frac{1}{2} \times r \times AC$$

However, the area of $\triangle ABC$ can also be expressed as

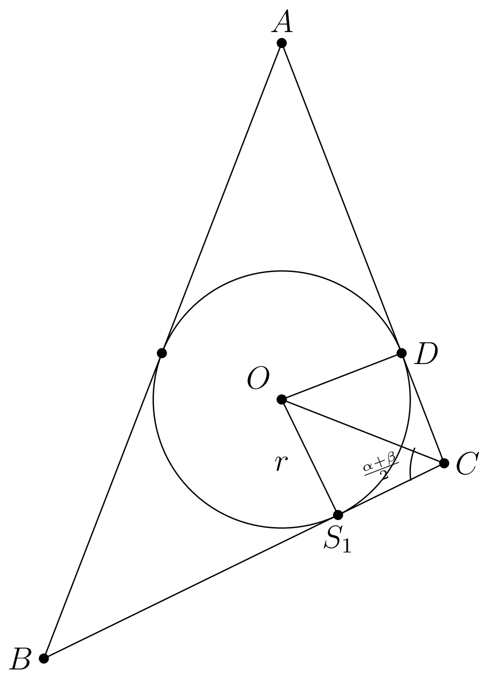
$$\text{Area} = \frac{1}{2} \times AB \times AC \times \sin(\pi - 2\alpha)$$

Equating the two expressions for the area gives

$$\frac{1}{2} \times r \times (AB + BC + AC) = \frac{1}{2} \times AC \times AC \times \sin 2\alpha$$

$$\therefore r = \frac{AC \times AB \sin 2\alpha}{AB + BC + AC}$$

(iv) Let O be the centre of the circle inscribed in $\triangle ABC$ and D be the point of contact of the circle to AC.



$$\begin{aligned}
OS_1 &= OD \quad \text{(equal radii)} \\
OC &\text{ is common} \\
S_1C &= DC \quad \text{(tangents from an external point are equal)} \\
\therefore \triangle OS_1C &\equiv \triangle ODC \quad (SSS) \\
\Rightarrow \angle OCD &= \angle OCS_1 \quad \text{(corresponding angles of congruent triangles)} \\
\Rightarrow \angle OCS_1 &= \frac{\alpha + \beta}{2} \\
\tan\left(\frac{\alpha + \beta}{2}\right) &= \frac{r}{CS_1} \\
CS_1 &= r \cot\left(\frac{\alpha + \beta}{2}\right)
\end{aligned}$$

But from part (a), it was shown that the smaller sphere touching the sides of the cone touches a focus of the ellipse. This means that circle touching the sides of $\triangle ABC$ touches BC at a focus of the ellipse, which is S_1. Hence

$$\begin{aligned} CS_1 &= BC - BS_1 \\ &= 2a - (a + ae) \\ &= a - ae \end{aligned}$$

Equating the two expressions for CS_1

$$\begin{aligned} a - ae &= r \cot\left(\frac{\alpha+\beta}{2}\right) \\ e &= 1 - \frac{r}{a}\cot\left(\frac{\alpha+\beta}{2}\right) \end{aligned}$$

(v) Starting with the results from (i) and (ii)

$$\begin{aligned} AC \times AB \sin 2\alpha &= \frac{4a^2\sin(\alpha+\beta)\sin(\alpha-\beta)}{\sin 2\alpha} \\ &= \frac{4a^2(\sin^2\alpha\cos^2\beta - \sin^2\beta\cos^2\alpha)}{2\sin\alpha\cos\beta} \\ &= \frac{2a^2(\sin^2\alpha(1-\sin^2\beta) - \sin^2\beta(1-\sin^2\alpha))}{\sin\alpha\cos\alpha} \\ &= \frac{2a^2(\sin^2\alpha - \sin^2\beta)}{\sin\alpha\cos\alpha} \\ &= \frac{2a^2(\sin\alpha+\sin\beta)(\sin\alpha-\sin\beta)}{\sin\alpha\cos\alpha} \\ AB + BC + AC &= \frac{2a\sin(\alpha+\beta)}{\sin 2\alpha} + 2a + \frac{2a\sin(\alpha-\beta)}{\sin 2\alpha} \\ &= \frac{2a(\sin 2\alpha + \sin(\alpha+\beta) + \sin(\alpha-\beta))}{\sin 2\alpha} \\ &= \frac{2a(2\sin\alpha\cos\alpha + 2\sin\alpha\cos\beta)}{2\sin\alpha\cos\alpha} \\ &= \frac{2a\sin\alpha(\cos\alpha+\cos\beta)}{\sin\alpha\cos\alpha} \end{aligned}$$

Substitute this into the result in (iii)

$$\begin{aligned}
r &= \frac{AC \times AB \sin 2\alpha}{AB + BC + AC} \\
&= \frac{a(\sin\alpha + \sin\beta)(\sin\alpha - \sin\beta)}{\sin\alpha(\cos\alpha + \cos\beta)} \\
\frac{r}{a} &= \frac{\sin\alpha + \sin\beta}{\cos\alpha + \cos\beta} \times \frac{\sin\alpha - \sin\beta}{\sin\alpha} \\
&= \frac{2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)}{2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)} \times \frac{\sin\alpha - \sin\beta}{\sin\alpha} \\
&= \tan\left(\frac{\alpha+\beta}{2}\right) \times \frac{\sin\alpha - \sin\beta}{\sin\alpha}
\end{aligned}$$

Substitute this into the expression in (iv)

$$\begin{aligned}
e &= 1 - \frac{r}{a}\cot\left(\frac{\alpha+\beta}{2}\right) \\
&= 1 - \tan\left(\frac{\alpha+\beta}{2}\right) \times \frac{\sin\alpha - \sin\beta}{\sin\alpha} \times \cot\left(\frac{\alpha+\beta}{2}\right) \\
&= 1 - \frac{\sin\alpha - \sin\beta}{\sin\alpha} \\
&= \frac{\sin\alpha - \sin\alpha + \sin\beta}{\sin\alpha} \\
&= \frac{\sin\beta}{\sin\alpha}
\end{aligned}$$

(c) (i) Since the game ends with either student A or student B winning then

$$P(A \text{ wins}) + P(B \text{ wins}) = 1$$

Since the game is perfectly symmetric then $P(A \text{ wins}) = P(B \text{ wins})$ so $P(A \text{ wins}) = \frac{1}{2}$.

(ii) If the first toss is a heads then student A is ahead of student B by one head. Consider the two cases going forward.

Case 1 - Second toss is a heads

One way for student A to win is for the third toss to also be heads (which gives HH as the next two outcomes). The probability that the second and third toss is heads is $\frac{1}{2} \times \frac{1}{2}$, or equivalently $\frac{1}{4}$.

If the third toss was tails instead, this 'cancels out' the second toss of heads and student A is back to being ahead of student B by one head. The probability of student A eventually winning after that is p by definition. Hence, the probability of student A to win after tossing HT as the next two outcomes is $\frac{1}{2} \times \frac{1}{2} \times p$, or equivalently $\frac{1}{4} \times p$.

Putting the two possibilities together, the probability that student A eventually wins if the first two tosses was heads is $\frac{1}{4} + \frac{1}{4} \times p$.

Case 2 - Second toss is a tails

This 'cancels out' the first toss of heads and student A is back to being tied with student B as if the game was to reset. The probability of student A eventually winning is $\frac{1}{2}$ as explained in part (i).

Hence, the probability that student A eventually wins after tossing T as the next outcome is $\frac{1}{2} \times \frac{1}{2}$, or equivalently $\frac{1}{4}$.

Putting all two cases together, the probability that student A wins the game after the first toss was a heads is given by

$$p = \frac{1}{2} + \frac{1}{4} \times p$$

(iii) Let q be the probability that student B wins if the first toss was a tails. By symmetry of the game, q must also satisfy the condition in (ii)

$$q = \frac{1}{2} + \frac{1}{4} \times q$$

This gives $q = \frac{2}{3}$.

The event that student A wins instead of student B, if the first toss was a tails is the complementary event. This has the probability of $\frac{1}{3}$.